

Modular Localization, Majorana Bilinears, and Near Saturation of the Tsirelson Bound

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Abstract

We investigate Bell–CHSH correlations generated by dichotomic observables constructed from bilinears of smeared Majorana fields. In contrast with constructions based directly on normalized Majorana fields, the observables considered here are even fermionic composites. We formulate their algebraic normalization and anticommutation requirements, construct modularly localized test functions using Tomita–Takesaki modular operators, and implement the required real-orthogonality conditions through a symplectic Gram–Schmidt procedure. The Bell correlators are evaluated through Wick contractions. A finite-dimensional numerical realization based on six Hermite–Gaussian modes, with Gaussian width σ , localized around modular frequency $\omega = 0$ yields, for $\sigma = 10^{-3}$,

$$\mathcal{B}_{\text{CHSH}} = 2.828406911727749,$$

corresponding to approximately 99.9993% of the Tsirelson value $2\sqrt{2}$. The four individual correlations simultaneously approach the ideal CHSH pattern.

1 Introduction

Bell correlations in relativistic quantum field theory provide a sharp meeting point between locality, entanglement, and the operator-algebraic structure of local observables. The two foundational papers of Summers and Werner develop the Bell framework for local quantum field theory and prove maximal vacuum violation for suitable spacelike separated observables in free Bose and Fermi theories [1, 2]. Their complementary-wedge analysis further shows that maximal violation is generic under the standard algebraic hypotheses of quantum field theory [3]. These results provide the natural point of departure for studying the Tsirelson bound with a different class of local observables.

The present work considers a different class of observables. Rather than using normalized smeared Majorana fields themselves as the dichotomic observables, we construct even operators from *bilinears* of Majorana fields. These operators belong to the bosonic even sector of the fermionic algebra. Their algebraic construction is consequently different, and their composite character also makes them potentially closer to parity-like and bilinear observables encountered in effective descriptions of Majorana degrees of freedom.

We work with a massive Majorana field in $1 + 1$ -dimensional Minkowski spacetime and with the standard right and left wedge geometry. The modular structure of wedge algebras is controlled by the Bisognano–Wichmann theorem [4, 5], which identifies the modular group with

Lorentz boosts. This allows the Tomita–Takesaki operators to be written particularly simply in rapidity space and, after Fourier transformation, in modular-frequency space.

A distinguished role is played by modular frequency $\omega = 0$. If δ denotes the modular operator, its spectral value in the ω representation is

$$\lambda(\omega) = e^{-2\pi\omega}. \quad (1)$$

Hence $\omega = 0$ corresponds precisely to the modular spectral value $\lambda = 1$. In the local quantum-field-theoretic setting this spectral structure is tied to the type-III₁ character of local von Neumann algebras. The value $\lambda = 1$ is fixed by the modular flow and is therefore a distinguished point of the modular spectrum. In the Summers–Werner analysis this point plays a special role in the mechanism leading to maximal Bell correlations. This observation gives the central motivation for concentrating our construction near $\omega = 0$: in the Fourier-dual boost representation, $\omega = 0$ is exactly the spectral representative of $\lambda = 1$. Numerically, we probe this region with Hermite–Gaussian wave packets of Gaussian width σ and study the approach to the Tsirelson value as σ decreases.

The paper is organized as follows. Section 2 introduces the massive 1 + 1 dimensional Majorana field in rapidity space, defines the bilinear observables, and establishes their Hermiticity and dichotomicity conditions. Section 3 reviews the Bisognano–Wichmann realization of Tomita–Takesaki modular theory for wedges, passes from rapidity to modular-frequency space, constructs Alice’s and Bob’s localized test functions, and implements the required real-orthogonality constraints by a symplectic Gram–Schmidt procedure. Section 4 derives the Bell–CHSH correlators using Wick contractions of smeared Majorana fields. Section 5 presents the Hermite–Gaussian finite-basis implementation and the numerical results. Section 6 discusses possible connections with effective Majorana modes in condensed-matter systems and directions for extending the modular construction. Finally, Appendix A provides the numerical implementation details, quadrature-convergence tests, high-precision certification, and reproducibility checks supporting the results quoted in the main text.

2 Majorana Bilinears as Dichotomic Observables

2.1 Massive Majorana field and one-particle Hilbert space

We consider a free massive Majorana field in 1 + 1 dimensions in wedge regions, following the modular framework developed in Ref. [8]. On the positive-energy mass shell, momentum is parametrized by the rapidity $\theta \in \mathbb{R}$ as

$$p^0(\theta) = m \cosh \theta, \quad p^1(\theta) = m \sinh \theta, \quad (2)$$

where units $c = 1$ are used. The one-particle Hilbert space is

$$\mathcal{H}_1 = L^2(\mathbb{R}, d\theta), \quad (3)$$

and for $f, g \in \mathcal{H}_1$ the Lorentz-invariant one-particle scalar product in rapidity variables is

$$\langle f, g \rangle = \int_{-\infty}^{\infty} d\theta \overline{f(\theta)} g(\theta). \quad (4)$$

The use of rapidity makes Lorentz boosts act as translations of θ , a fact that will be essential for the modular construction below.

Let $\psi(f)$ denote the Majorana field smeared with $f \in \mathcal{H}_1$. We use the CAR normalization

$$\{\psi(f), \psi(g)\} = 2 \Re \langle f, g \rangle, \quad (5)$$

so that

$$\psi(f)^2 = \|f\|^2. \quad (6)$$

2.2 Alice's and Bob's bilinears

For two one-particle vectors $f, h \in \mathcal{H}_1$, define

$$A = -i \frac{\psi(h)\psi(f)}{\|h\| \|f\|}, \quad A' = -i \frac{\psi(h')\psi(f')}{\|h'\| \|f'\|}. \quad (7)$$

We require

$$\Re\langle h, f \rangle = 0, \quad \Re\langle h', f' \rangle = 0. \quad (8)$$

By Eq. (5), these conditions imply that the two Majorana factors inside each bilinear anticommute. Consequently,

$$A^\dagger = A, \quad (A')^\dagger = A', \quad (9)$$

and

$$A^2 = (A')^2 = \mathbf{1}. \quad (10)$$

Thus A and A' are Hermitian dichotomic observables.

For Bob we use the notation

$$B = -i \frac{\psi(p)\psi(g)}{\|p\| \|g\|}, \quad B' = -i \frac{\psi(p')\psi(g')}{\|p'\| \|g'\|}, \quad (11)$$

with

$$\Re\langle g, p \rangle = 0, \quad \Re\langle g', p' \rangle = 0. \quad (12)$$

The same argument gives $B^\dagger = B$, $(B')^\dagger = B'$ and $B^2 = (B')^2 = \mathbf{1}$. No additional condition requiring, for example, $\psi(g)$ and $\psi(g')$ to anticommute is imposed.

3 Tomita–Takesaki Modular Localization in Wedges

3.1 Bisognano–Wichmann theorem in rapidity space

For the right wedge, the Bisognano–Wichmann theorem identifies the modular flow with Lorentz boosts [4, 5]. In rapidity space the one-particle boost generator is

$$k = -i \frac{d}{d\theta}, \quad (13)$$

and, with the conventions used here, the modular operator is

$$\delta = e^{-2\pi k}. \quad (14)$$

Therefore

$$(\delta^{1/2}f)(\theta) = f(\theta + i\pi). \quad (15)$$

For the massive Majorana one-particle space we use the modular conjugation convention

$$(jf)(\theta) = \overline{f(\theta)}, \quad (16)$$

and the Tomita–Takesaki antilinear operator is

$$s = j\delta^{1/2}. \quad (17)$$

The real standard subspace associated with the wedge is characterized by the fixed-point condition $sf = f$.

3.2 Dual modular-frequency representation

We now pass to the Fourier-dual variable ω , which diagonalizes the boost generator. Our convention is

$$f(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} d\theta e^{-i\omega\theta} f(\theta), \quad (18)$$

with

$$\langle f, g \rangle = \int_{-\infty}^{\infty} d\omega \overline{f(\omega)} g(\omega). \quad (19)$$

In this representation,

$$(\delta f)(\omega) = e^{-2\pi\omega} f(\omega), \quad (20)$$

$$(\delta^{1/2} f)(\omega) = e^{-\pi\omega} f(\omega), \quad (21)$$

$$(j f)(\omega) = \overline{f(-\omega)}, \quad (22)$$

$$(s f)(\omega) = e^{\pi\omega} \overline{f(-\omega)}. \quad (23)$$

The adjoint Tomita–Takesaki map acts as

$$(s^\dagger f)(\omega) = e^{-\pi\omega} \overline{f(-\omega)}. \quad (24)$$

The modular spectral value is $\lambda(\omega) = e^{-2\pi\omega}$; hence $\omega = 0$ is exactly the point $\lambda = 1$.

3.3 Modular localization, standard subspaces, and wedge duality

We briefly make explicit what is meant here by modular localization; see, for example, the review by Guido [7]. Let

$$\mathcal{K}(W_R) = \{f \in \text{Dom}(s) : s f = f\} \quad (25)$$

be the real standard one-particle subspace associated with the right wedge. Standardness means

$$\mathcal{K}(W_R) \cap i\mathcal{K}(W_R) = \{0\}, \quad \overline{\mathcal{K}(W_R) + i\mathcal{K}(W_R)} = \mathcal{H}_1. \quad (26)$$

The Tomita–Takesaki data (s, j, δ) are the one-particle modular objects associated with this standard subspace.

The projector-like real-linear map $1 + s$ maps suitable vectors in $\text{Dom}(s)$ into the fixed-point space $\mathcal{K}(W_R)$ and is therefore the basic localization map used for Alice. The complementary left wedge is described by the corresponding dual standard subspace. In the fermionic theory ordinary Haag duality is replaced by *twisted duality*; with the phase convention adopted below, Bob's vectors are generated from $1 + s^\dagger$ together with the required factors of i .

With the present CAR convention, right- and twisted-left-localized one-particle vectors are real-orthogonal,

$$\Re \langle f, g \rangle = 0, \quad f \in \mathcal{K}(W_R), \quad g \in \mathcal{K}_{\text{tw}}(W_L). \quad (27)$$

Consequently the corresponding odd Majorana fields have the required graded locality. Since the observables A, A', B, B' are even bilinears, the twist disappears at the observable level and complementary-wedge bilinears commute. In particular,

$$[A, B] = [A, B'] = [A', B] = [A', B'] = 0. \quad (28)$$

This is the modular-localization origin of the Alice–Bob commutativity required by the Bell construction.

3.4 Alice's modularly localized test functions

Let $\varphi(\omega)$ be real. Alice's first pair is generated by

$$f = (1 + s)\varphi, \quad (29)$$

$$f' = (1 + s)i\varphi. \quad (30)$$

Using Eq. (23), these become explicitly

$$f(\omega) = \varphi(\omega) + e^{\pi\omega}\varphi(-\omega), \quad (31)$$

$$f'(\omega) = i[\varphi(\omega) - e^{\pi\omega}\varphi(-\omega)]. \quad (32)$$

For the second Majorana factor we take real functions ℓ, η, ℓ', η' and define

$$h = (1 + s)(\ell + i\eta), \quad (33)$$

$$h' = (1 + s)(\ell' + i\eta'). \quad (34)$$

Thus

$$h(\omega) = \ell(\omega) + i\eta(\omega) + e^{\pi\omega}[\ell(-\omega) - i\eta(-\omega)], \quad (35)$$

$$h'(\omega) = \ell'(\omega) + i\eta'(\omega) + e^{\pi\omega}[\ell'(-\omega) - i\eta'(-\omega)]. \quad (36)$$

The required conditions are $\Re\langle f, h \rangle = 0$ and $\Re\langle f', h' \rangle = 0$.

3.5 Bob, the commutant, and twisted duality

Bob's vectors are constructed from the dual Tomita–Takesaki map $s^\dagger$. The factors of i are essential: they implement the fermionic twist required by twisted duality. let $\widehat{g}(\omega)$ be real and define

$$g = i(1 + s^\dagger)\widehat{g}, \quad (37)$$

$$g' = -i(1 + s^\dagger)i\widehat{g}. \quad (38)$$

Hence

$$g(\omega) = i[\widehat{g}(\omega) + e^{-\pi\omega}\widehat{g}(-\omega)], \quad (39)$$

$$g'(\omega) = \widehat{g}(\omega) - e^{-\pi\omega}\widehat{g}(-\omega). \quad (40)$$

For the second factor of Bob's bilinears, with real $\alpha, \beta, \alpha', \beta'$, we set

$$p = i(1 + s^\dagger)(\alpha + i\beta), \quad (41)$$

$$p' = -i(1 + s^\dagger)(\alpha' + i\beta'). \quad (42)$$

Equivalently,

$$p(\omega) = i\left\{\alpha(\omega) + i\beta(\omega) + e^{-\pi\omega}[\alpha(-\omega) - i\beta(-\omega)]\right\}, \quad (43)$$

$$p'(\omega) = -i\left\{\alpha'(\omega) + i\beta'(\omega) + e^{-\pi\omega}[\alpha'(-\omega) - i\beta'(-\omega)]\right\}. \quad (44)$$

The internal Bob constraints are $\Re\langle g, p \rangle = 0$ and $\Re\langle g', p' \rangle = 0$.

3.6 Symplectic Gram–Schmidt procedure

The orthogonality relevant to the fermionic bilinears is the real part of the one-particle scalar product. It is this real bilinear form that enters the CAR anticommutator in Eq. (5); accordingly, the Gram–Schmidt step used here is naturally adapted to the real/symplectic structure of the one-particle space rather than to complex Hilbert-space orthogonality.

For two generic vectors $u, v \in \mathcal{H}_1$, define

$$v_{\perp} = v - \frac{\Re\langle u, v \rangle}{\|u\|^2} u. \quad (45)$$

Then

$$\Re\langle u, v_{\perp} \rangle = 0. \quad (46)$$

We refer to this real projection as the symplectic Gram–Schmidt procedure. It is applied independently to the two Majorana factors entering each bilinear.

For example, for the specific Alice pair, one may define, for a suitable ξ ,

$$h_{\perp} = \xi - \frac{\Re\langle f, \xi \rangle}{\|f\|^2} f, \quad (47)$$

so that

$$\Re\langle f, h_{\perp} \rangle = 0 \implies \{\psi(f), \psi(h_{\perp})\} = 0. \quad (48)$$

The same operation is performed independently for (f', h') , (g, p) , and (g', p') . In the numerical implementation the projection is imposed before the Bell correlators are evaluated.

4 Bell–CHSH Correlators and Wick Contractions

We now return to the four Majorana bilinears A, A', B, B' constructed above. They satisfy the Bell algebra

$$A^2 = (A')^2 = B^2 = (B')^2 = \mathbf{1}, \quad (49)$$

$$[A, B] = [A, B'] = [A', B] = [A', B'] = 0, \quad (50)$$

where the first line follows from the internal CAR orthogonality of each bilinear and the second from modular localization in complementary wedges and fermionic twisted duality.

The CHSH operator is

$$\mathcal{C} = (A + A')B + (A - A')B'. \quad (51)$$

We denote its vacuum expectation value by

$$\mathcal{B}_{\text{CHSH}} := \langle \Omega, \mathcal{C}\Omega \rangle = \langle AB \rangle + \langle A'B \rangle + \langle AB' \rangle - \langle A'B' \rangle. \quad (52)$$

Here Ω denotes the vacuum vector. Thus $\mathcal{C}$ always denotes the CHSH operator, whereas $\mathcal{B}_{\text{CHSH}}$ denotes the corresponding Bell parameter in the vacuum state Ω .

Each Alice–Bob correlator is a four-point function of *smeared* Majorana fields. For example,

$$\langle AB \rangle = -\frac{\langle \psi(h)\psi(f)\psi(p)\psi(g) \rangle}{\|h\| \|f\| \|p\| \|g\|}. \quad (53)$$

More generally, for $F_1, F_2, F_3, F_4 \in \mathcal{H}_1$, quasifreeness gives the Wick identity

$$\begin{aligned} & \langle \psi(F_1)\psi(F_2)\psi(F_3)\psi(F_4) \rangle \\ &= \langle F_1, F_2 \rangle \langle F_3, F_4 \rangle - \langle F_1, F_3 \rangle \langle F_2, F_4 \rangle + \langle F_1, F_4 \rangle \langle F_2, F_3 \rangle, \end{aligned} \quad (54)$$

with the two-point convention inherited from the smeared Majorana field. In particular,

$$\langle AB \rangle = -\frac{1}{\|h\| \|f\| \|p\| \|g\|} \{ \langle h, f \rangle \langle p, g \rangle - \langle h, p \rangle \langle f, g \rangle + \langle h, g \rangle \langle f, p \rangle \}. \quad (55)$$

The other three correlations follow by the corresponding primed substitutions.

5 Hermite–Gaussian Approximation and Numerical Results

5.1 Finite trial space

For Gaussian width $\sigma > 0$, the numerical trial space is generated by six Hermite–Gaussian modes, $n = 0, \dots, 5$,

$$\chi_n^{(\sigma)}(\omega) = e^{-\omega^2/(2\sigma^2)} \frac{H_n(\omega/\sigma)}{\sqrt{2^n n!}}. \quad (56)$$

The modular seed functions introduced in the preceding section are expanded in this finite basis before the localization maps are applied. For example,

$$f = (1 + s)\varphi, \quad \varphi(\omega) = \sum_{n=0}^5 c_n \chi_n^{(\sigma)}(\omega), \quad (57)$$

with variational coefficients c_n . The analogous expansion is made for each of the remaining real seed functions entering f', h, h', g, g', p, p' . Their finite sets of coefficients are then optimized numerically, while the modular localization maps and the symplectic Gram–Schmidt constraints are imposed before the Bell correlators are evaluated.

The small- σ regime concentrates the spectral support near $\omega = 0$, i.e. near the modular spectral value $\lambda = 1$.

5.2 Convergence toward the Tsirelson value

The optimized values obtained in the six-mode trial space are

σ	$\mathcal{B}_{\text{CHSH}}$	$2\sqrt{2} - \mathcal{B}_{\text{CHSH}}$
1.0×10^{-2}	2.82474588	3.68×10^{-3}
5.0×10^{-3}	2.82718767	1.24×10^{-3}
2.0×10^{-3}	2.82820789	2.19×10^{-4}
1.0×10^{-3}	2.82840691	2.02×10^{-5}

For reference,

$$2\sqrt{2} = 2.828427124746\dots \quad (58)$$

At $\sigma = 10^{-3}$,

$$\mathcal{B}_{\text{CHSH}} = 2.828406911727749, \quad (59)$$

so that

$$\frac{\mathcal{B}_{\text{CHSH}}}{2\sqrt{2}} \simeq 0.999992853. \quad (60)$$

The finite six-mode trial space therefore reaches approximately 99.9993% of the Tsirelson value.

The four individual correlators at $\sigma = 10^{-3}$ are

Correlator	Numerical value	Tsirelson target
$\langle AB \rangle$	0.7069610194038407592	$+1/\sqrt{2}$
$\langle A'B \rangle$	0.7072492746151345647	$+1/\sqrt{2}$
$\langle AB' \rangle$	0.7070410913686358234	$+1/\sqrt{2}$
$\langle A'B' \rangle$	-0.7071555263401378480	$-1/\sqrt{2}$

Thus the four entries approach the ideal CHSH pattern $(1/\sqrt{2}, 1/\sqrt{2}, 1/\sqrt{2}, -1/\sqrt{2})$ individually, providing a stronger diagnostic than the CHSH sum alone.

With the optimized coefficients held fixed, the $\sigma = 10^{-3}$ result is stable under Gauss–Hermite quadratures with 80, 120, 180, and 240 points, yielding

$$\mathcal{B}_{\text{CHSH}} = 2.828406911727749 \quad (61)$$

at the displayed precision. Within this finite numerical experiment, the observed trend is consistent with

$$\sigma \rightarrow 0 \implies \omega \rightarrow 0 \implies \lambda \rightarrow 1 \implies \mathcal{B}_{\text{CHSH}} \rightarrow 2\sqrt{2}. \quad (62)$$

This is numerical evidence for near saturation and not, by itself, an analytic proof of the limiting value.

6 Condensed-Matter Perspective and Outlook

The use of even Majorana bilinears suggests a possible conceptual bridge between the modular field-theoretic construction above and effective Majorana degrees of freedom in condensed-matter systems. Majorana zero modes are predicted in topological superconducting settings and have been extensively studied in atomic spin chains on superconducting substrates and related hybrid structures; for a recent comprehensive review, see Ref. [10].

A possible experimental program would seek effective observables playing the roles of A, A', B, B' while preserving the algebraic ingredients responsible for the large CHSH correlations: dichotomicity, controlled anticommutation within each bilinear, sufficiently independent measurement settings, and an effective analogue of the localization structure used in the modular construction. Because the observables studied here are even Majorana bilinears, parity-sensitive measurements and couplings between effective Majorana modes are natural objects to investigate.

This connection is presently a perspective rather than a microscopic experimental proposal. A concrete realization would require identifying experimentally accessible bilinear/parity measurements, mapping the modularly localized one-particle test functions to controllable effective modes, and analyzing finite temperature, quasiparticle poisoning, decoherence, detector locality, and measurement compatibility.

The present work has deliberately focused on a massive Majorana field in $1+1$ dimensions and wedge localization, where the Bisognano–Wichmann theorem gives direct control of the modular data. A natural extension is to investigate massless/chiral Majorana fields on light rays and localization in bounded regions such as double cones. In that broader setting, modular localization methods of the Brunetti–Guido–Longo type may provide the appropriate framework [9]. This extension is left for future work.

7 Conclusions

We have constructed Hermitian dichotomic observables from modularly localized Majorana bilinears in the wedge geometry of a massive $1+1$ -dimensional Majorana field. The construction makes explicit the role of the CAR real-orthogonality conditions, the Tomita–Takesaki modular maps, twisted duality for the complementary wedge, and the real Gram–Schmidt projection used to enforce the bilinear constraints.

The finite six-mode Hermite–Gaussian implementation approaches the Tsirelson value with high numerical precision as the Gaussian width is reduced and the trial functions become concentrated near modular frequency $\omega = 0$, corresponding to the distinguished modular spectral value $\lambda = 1$. At $\sigma = 10^{-3}$ the reconstructed optimum gives $\mathcal{B}_{\text{CHSH}} = 2.828406911727749$ at the displayed precision, corresponding to approximately 99.9993% of $2\sqrt{2}$, while all four individual correlators simultaneously approach the ideal Tsirelson pattern. Appendix ?? documents the associated quadrature-convergence tests, the 60-digit reevaluation of the optimized point, and the numerical verification of the CAR real-orthogonality constraints.

An analytic control of the $\sigma \rightarrow 0$ limit remains an important open problem. Further directions include enlarging the trial space, studying the robustness of the optimizing configurations, extending the modular construction beyond wedges, and investigating whether effective condensed-matter Majorana platforms can realize useful analogues of the bilinear observables considered

here. Another natural direction is to generalize the present construction beyond bilinears to higher-order Majorana composites, including trilinear, quadrilinear, and more general multilinear operators. Such extensions may lead to new classes of Bell observables and to a richer hierarchy of quantum correlations within the same modular framework, and will be investigated in future work.

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Data Availability

The numerical data supporting the findings of this study are contained within the article. Additional numerical details can be made available by the authors upon reasonable request.

A Numerical Implementation and Precision Analysis

This appendix summarizes the numerical implementation used for the finite Hermite–Gaussian realization of the modular Majorana-bilinear Bell–CHSH construction and records the corresponding precision checks. The aim is to separate clearly the variational search, quadrature convergence, arbitrary-precision reevaluation, and enforcement of the real CAR-orthogonality constraints.

A.1 Finite Hermite–Gaussian trial space

Let ε denote the modular-frequency variable. For a Gaussian width $\sigma > 0$, the numerical seed functions are expanded in the first six Hermite–Gaussian modes,

$$\Phi_n^{(\sigma)}(\varepsilon) = e^{-\varepsilon^2/(2\sigma^2)} \frac{H_n(\varepsilon/\sigma)}{\sqrt{2^n n!}}, \quad n = 0, \dots, 5. \quad (63)$$

Each real seed function entering the modular construction is written as

$$q(\varepsilon) = \sum_{n=0}^5 c_n \Phi_n^{(\sigma)}(\varepsilon), \quad (64)$$

with real variational coefficients c_n . The same six-dimensional ansatz is used independently for the seed functions from which $f, f', h, h', g, g', p, p'$ are generated.

The small- σ regime concentrates the spectral support near $\varepsilon = 0$. In the modular spectral parametrization employed in the main text, this is the region corresponding to the distinguished modular spectral value $\lambda = 1$.

A.2 Modular localization before optimization

For every trial point, the modular localization maps are applied before the Bell correlators are evaluated. Alice’s vectors are generated with $1 + s$, whereas Bob’s vectors are generated with $1 + s^\dagger$, including the fermionic twist phases used in the main construction. Thus the variational coefficients belong to the seed functions, while the objective function is evaluated on the corresponding modularly localized vectors.

The numerical ordering is therefore

$$\text{seed coefficients} \longrightarrow \text{modularly localized vectors} \longrightarrow \text{CAR projection} \longrightarrow \mathcal{B}_{\text{CHSH}}. \quad (65)$$

In particular, the routine does not optimize unconstrained Bell correlators and only afterwards test localization.

A.3 Symplectic Gram–Schmidt projection and CAR constraints

For a pair of one-particle vectors u, v , the relevant fermionic constraint is

$$\Re\langle u, v \rangle = 0. \quad (66)$$

The numerical projection is the real, or symplectic, Gram–Schmidt step

$$v_{\perp} = v - \frac{\Re\langle u, v \rangle}{\|u\|^2} u. \quad (67)$$

It is applied independently to

$$(f, h), \quad (f', h'), \quad (g, p), \quad (g', p'). \quad (68)$$

At the final high-precision reevaluation, the residual real-orthogonality constraints were found to be

$$\Re\langle f, h \rangle \sim 7.1 \times 10^{-62}, \quad (69)$$

$$\Re\langle f', h' \rangle \sim 2.8 \times 10^{-61}, \quad (70)$$

$$\Re\langle g, p \rangle \sim -3.0 \times 10^{-61}, \quad (71)$$

$$\Re\langle g', p' \rangle \sim 1.1 \times 10^{-62}. \quad (72)$$

Thus the internal CAR constraints remain satisfied essentially at the full working precision of the final reevaluation.

A.4 Evaluation of the Bell correlators

Each Bell correlator is a four-point vacuum expectation value of smeared Majorana fields. Quasifreeness reduces these quantities to one-particle scalar products through Wick’s theorem,

$$\begin{aligned} \langle \psi(F_1)\psi(F_2)\psi(F_3)\psi(F_4) \rangle &= \langle F_1, F_2 \rangle \langle F_3, F_4 \rangle - \langle F_1, F_3 \rangle \langle F_2, F_4 \rangle \\ &\quad + \langle F_1, F_4 \rangle \langle F_2, F_3 \rangle. \end{aligned} \quad (73)$$

The normalization factors associated with the Majorana bilinears are then included explicitly. The Bell parameter is

$$\mathcal{B}_{\text{CHSH}} = \langle AB \rangle + \langle A'B \rangle + \langle AB' \rangle - \langle A'B' \rangle. \quad (74)$$

Once the localized vectors have been constructed, the numerical problem thus reduces to repeated evaluation of one-dimensional scalar products in ε .

A.5 Gauss–Hermite quadrature

Because the basis functions in Eq. (63) carry a Gaussian factor, Gauss–Hermite quadrature provides a natural discretization of the required one-dimensional integrals. For $\sigma = 10^{-3}$, the final optimized point was reevaluated without changing the variational coefficients while increasing the quadrature order. The result is The displayed value is therefore stable to all 15 digits shown already from $N = 80$ through $N = 240$. This test controls the discretization of the integral and should be distinguished from the floating-point precision used in the variational search.

Gauss–Hermite order N	$\mathcal{B}_{\text{CHSH}}$
80	2.828406911727749
120	2.828406911727749
180	2.828406911727749
240	2.828406911727749

Table 1: Quadrature convergence at $\sigma = 10^{-3}$ for the reconstructed six-mode optimum.

A.6 Optimization precision and high-precision certification

The numerical workflow was carried out in two stages.

First, the variational optimization was performed in ordinary double precision. This stage is computationally efficient and is used to locate a high-quality point in the finite-dimensional coefficient space.

Second, the optimized coefficient vector was held fixed and all relevant scalar products, Gram–Schmidt projections, normalizations, Wick contractions, and the final CHSH combination were reevaluated using 60-digit arithmetic. The resulting value is

$$\mathcal{B}_{\text{CHSH}} = 2.8284069117277489952982694957984961868084126360042. \quad (75)$$

The residual imaginary part of the final Bell value is of order 10^{-61} and is therefore numerically zero at the working precision.

It is important to distinguish these two stages. The nonlinear optimization itself was not performed at 60-digit precision; rather, the optimum was located in double precision and subsequently reevaluated and numerically certified at 60 digits.

A.7 Distance from the Tsirelson bound

The exact Tsirelson value is

$$2\sqrt{2} = 2.8284271247461900976\dots \quad (76)$$

For the reconstructed six-mode optimum,

$$2\sqrt{2} - \mathcal{B}_{\text{CHSH}} \simeq 2.02130184411023 \times 10^{-5}, \quad (77)$$

or equivalently

$$\frac{\mathcal{B}_{\text{CHSH}}}{2\sqrt{2}} \simeq 0.999992853\dots \quad (78)$$

Thus the finite six-mode realization reaches approximately 99.9993% of the Tsirelson bound.

The four individual correlators at this point are

$$\langle AB \rangle = 0.7069610194038407592, \quad (79)$$

$$\langle A'B \rangle = 0.7072492746151345647, \quad (80)$$

$$\langle AB' \rangle = 0.7070410913686358234, \quad (81)$$

$$\langle A'B' \rangle = -0.7071555263401378480. \quad (82)$$

Each entry is close to the ideal Tsirelson pattern

$$\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right), \quad (83)$$

which provides a stronger diagnostic than the CHSH sum alone.

A.8 Scope of the numerical checks

The convergence and precision tests reported above establish that, for the reconstructed finite-dimensional optimum, the Bell value is insensitive to quadrature refinement and stable under arbitrary-precision reevaluation. They also verify that the internal CAR constraints are satisfied to extremely high numerical accuracy.

These checks do not constitute an analytic proof that

$$\mathcal{B}_{\text{CHSH}} \longrightarrow 2\sqrt{2} \quad \text{as} \quad \sigma \longrightarrow 0, \quad (84)$$

nor do they prove that the six-mode point is the global optimum of the full infinite-dimensional variational problem. Their appropriate interpretation is therefore as high-accuracy numerical evidence for near saturation of the Tsirelson bound within the finite Hermite–Gaussian realization.

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